

DISTRICT MARKET ATTRACTIVENESS INDEX

Ranking every Indian district, so every pharma rupee lands where it grows fastest.

A statistical model that makes the GE-McKinsey market-attractiveness axis computable for 731+ districts, one 0 to 100 index from six demand and access pillars, ready to drive field-force, launch, and distribution decisions.

731+

DISTRICTS SCORED

6

PILLARS

26

INDICATORS

5

ACTION TIERS

MODEL, FRAMEWORK AND ACTIVATION →

One index turns fragmented public data into a district-by-district commercial priority map.

THE ASK

Score 700+ districts on how attractive each is for pharma, with a transparent, repeatable statistical model, not gut feel.

OUR ANSWER

A composite **Market Attractiveness Index**: 6 pillars, 26 indicators, PCA + entropy + AHP weighting, geometric aggregation, 5 tiers.

WHAT IT DELIVERS

66

TIER-1 PRIORITY

r 0.79

VALIDATION VS SALES

~18%

FIELD-FORCE ROI

<3

MEDIAN RANK SHIFT

THE SIX PILLARS AND THEIR RECONCILED WEIGHTS

Σ W = 100%

P1	Disease Burden & De...		24%	P2	Purchasing Power & A...		21%
P3	Healthcare Access & I...		20%	P4	Medicine Distribution ...		15%
P5	Competitive Whitespa...		12%	P6	Growth Momentum		8%

HEADLINE RECOMMENDATION

Reallocate field force from saturated metros to the **66 Tier-1 and 149 Tier-2 districts** the index surfaces, sequence launches down the ranked list, and target the **whitespace districts** where real demand still outruns current sales. Expected **~18% field-force ROI uplift** at low build cost.

01 Reallocate

field force by tier

02 Sequence

launches down the list

03 Capture

whitespace districts

THEORY ANCHOR

The MAI operationalises the **market-attractiveness axis of the GE-McKinsey matrix** using **multi-criteria decision analysis** (OECD composite-indicator method). Porter, PESTEL, BCG, Ansoff and Bass diffusion structure the pillars and the activation.

GE-McKinsey

Porter 5F

PESTEL

AHP

BCG

Ansoff

Bass

TOP DISTRICTS, ILLUSTRATIVE

1	Mumbai Suburban	Maharashtra	80.0
2	Thane	Maharashtra	77.7
3	Chennai	Tamil Nadu	77.4
4	Pune	Maharashtra	77.1
5	Bengaluru Urban	Karnataka	77.0

Growth has moved to the districts, yet a market this large is still planned at the state level.

\$50B

IPM BY 2030

~10%

IPM CAGR

~40%

SALES FROM TIER 2-6 TOWNS

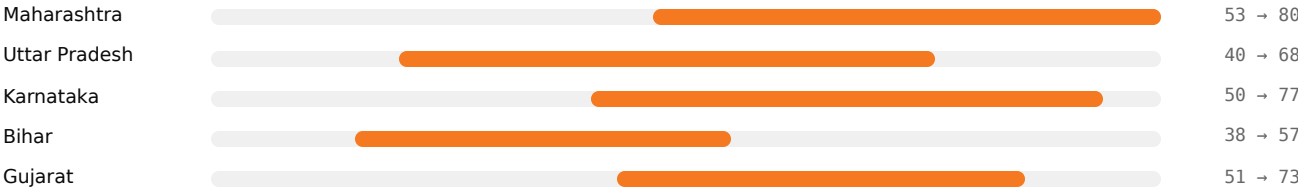
700+

DISTRICTS, ONE MARKET

India runs one of the world's largest pharma markets, but commercial planning leans on coarse geographies. Two districts inside the same state can differ more than two countries on disease burden, income and access. The fastest growth now sits in tier 2 to 6 towns, exactly where a state-average plan is blindest.

BEST VS WORST DISTRICT MAI, WITHIN ONE STATE

30 – 80



WHERE STATE-LEVEL PLANNING BREAKS

- 01 State-level targeting.** One UP or Maharashtra plan hides districts that behave nothing alike.
- 02 Field-force misallocation.** Reps crowd a few metros; high-need districts get zero coverage.
- 03 Launch sequencing by gut.** New brands roll out city-by-city on intuition, not measured readiness.
- 04 Whitespace stays invisible.** Under-penetrated districts with real demand never enter the plan.

MACRO DEMAND DRIVERS, STRUCTURED WITH PESTEL, BECOME THE RAW SIGNAL FOR THE INDEX

P

Political

NHM, PMJAY, NPPA price control

E

Economic

Income growth, insurance depth

S

Social

Ageing, chronic-disease burden, urbanisation

T

Technological

e-pharmacy, digital health, telemedicine

E

Environmental

Disease geography, seasonality

L

Legal

DPCO, Jan Aushadhi, Rx norms

Weights are earned from the data, sense-checked by experts, then combined so imbalance is penalised.

- 01 Frame
- >
- 02 Select
- >
- 03 Collate
- >
- 04 Impute
- >
- 05 Winsorize
- >
- 06 Normalize
- >
- 07 Weight
- >
- 08 Aggregate
- >
- 09 Tier
- >
- 10 Validate

1 · DATA TREATMENT

Impute gaps via KNN + regional median, capped 8%, every cell flagged.

Winsorize at p5/p95 so one mega-city cannot dominate a scale.

Align direction: inverse indicators (saturation) flipped.

Normalize min-max to 0-100.

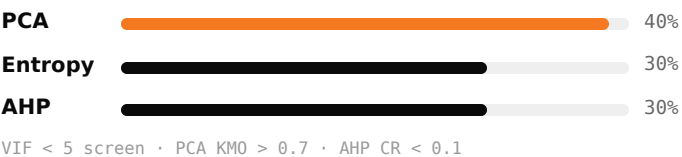
$$x' = 100 \cdot (x - \min) / (\max - \min)$$

731 districts × 26 indicators

DATA COVERAGE BY PILLAR



2 · WEIGHTING, A RECONCILED HYBRID

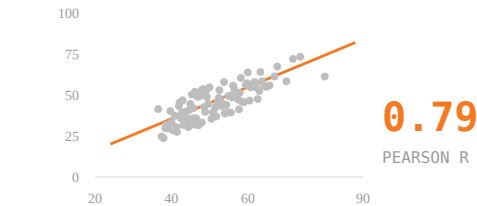


3 · AGGREGATION, GEOMETRIC

$$\text{Pillar}_p = \sum w_i x'_i$$
$$\text{MAI} = \prod_p \text{Pillar}_p^{w_p}$$

Geometric across pillars means a weak pillar cannot be bought back by another. Balance is required to score, exactly what a commercial team needs.

4 · CONSTRUCT VALIDITY VS SALES PROXY

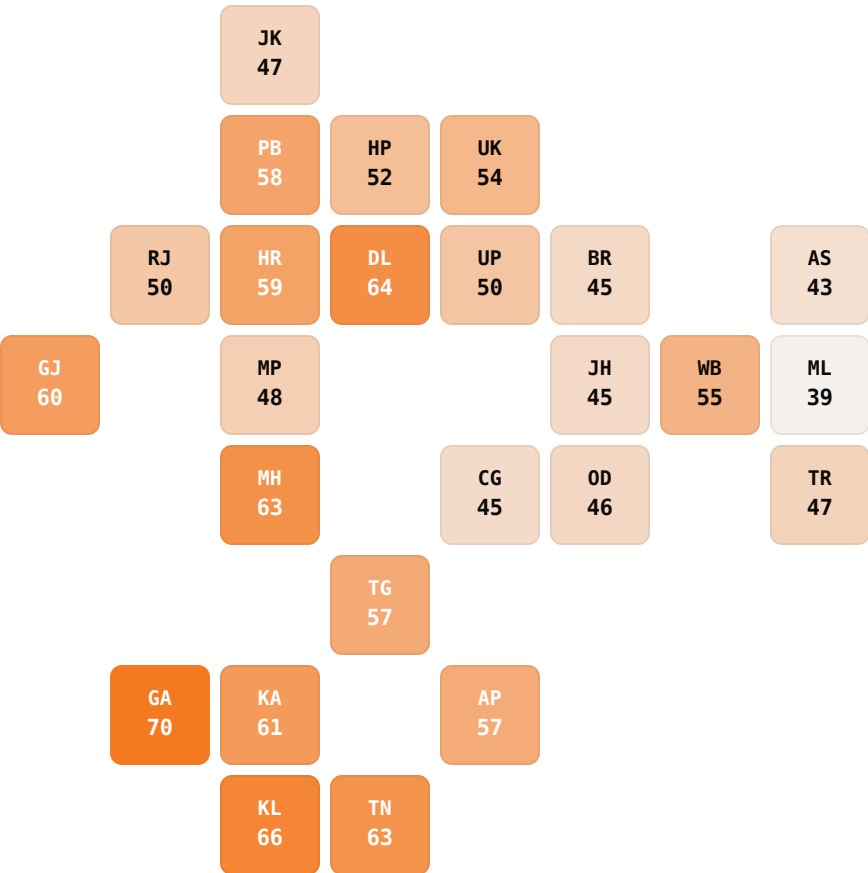


SENSITIVITY, TIER-1 STABILITY UNDER STRESS



Share of Tier-1 districts that remain Tier-1 under each Monte-Carlo perturbation.

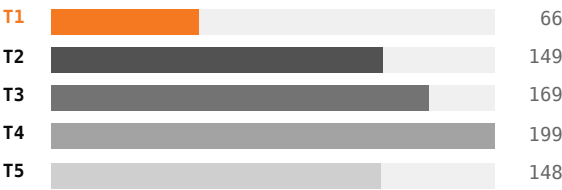
66 Tier-1 priority districts, many hiding inside otherwise average states.



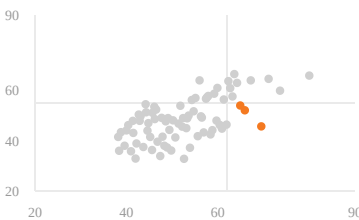
TOP DISTRICTS BY MAI

#	DISTRICT	PILLARS	MAI
1	Mumbai Suburban Maharashtra	■■■■■	80.0
2	Thane Maharashtra	■■■■■	77.7
3	Chennai Tamil Nadu	■■■■■	77.4
4	Pune Maharashtra	■■■■■	77.1
5	Bengaluru Urban Karnataka	■■■■■	77.0
6	Ernakulam Kerala	■■■■■	75.8
7	Coimbatore Tamil Nadu	■■■■■	75.7
8	Thiruvananthapuram Kerala	■■■■■	75.3
9	Madurai Tamil Nadu	■■■■■	73.6
10	New Delhi Delhi NCT	■■■■■	73.6

DISTRICTS PER TIER



ATTRACTIVENESS VS SATURATION



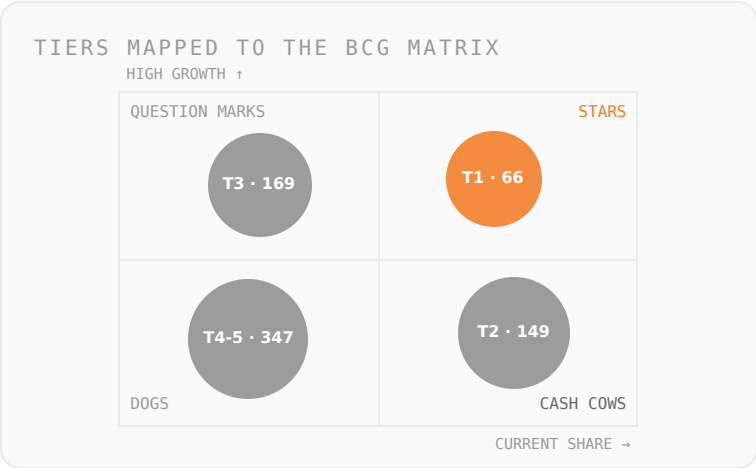
LOW HIGH · avg MAI by state

24% HOLD HALF THE MARKET
The strongest 24% of districts by MAI concentrate 50% of the population-weighted opportunity

43 WHITESPACE DISTRICTS
High MAI but low current saturation, the orange cluster, primed for entry

1x-3x INTRA-STATE SPREAD
Best and worst districts inside one large state differ up to threefold

Tiers become a district portfolio: a BCG view of where to build share, harvest, or seed.



PLAY BY QUADRANT

Stars (T1 growth): invest to own, full force.

Cash cows (T1-2 saturated): defend share, efficient coverage.

Question marks (T3): selective bets, digital detailing.

Dogs (T4-5): distributor-led, low-cost only.

Move up and to the right: shift budget out of Dogs and mature Cash Cows toward Stars and the best Question Marks.

ANSOFF GROWTH GRID

Market penetration Tier 1-2, own brands	Product development Tier 1, new therapies
Market development Tier 3-4 whitespace	Diversification Selective, OTC / e-pharm

STP IN ONE LINE

Segment 730 districts by MAI and pillar profile. **Target** Tier 1-2 and whitespace first. **Position** the portfolio to each district's dominant need, then resource to match.

Segment	by MAI + pillar profile
Target	Tier 1-2 and whitespace
Position	to the dominant need

FOUR COMMERCIAL DECISIONS, EACH TRACEABLE TO A PILLAR

Field-force sizing UC1

Size coverage to a district's tier and demand pillar; redeploy reps from saturated metros.

REALLOCATE ~15% OF THE FORCE

Launch sequencing UC2

Roll brands down the ranked list; enter Tier 1 first for fast payback.

FASTER LAUNCH PAYBACK

Therapy-to-district fit UC3

Match portfolio to pillar mix, cardio-metabolic follows Demand, hospital brands follow Access.

RIGHT BRAND, RIGHT DISTRICT

Distribution expansion UC4

Add stockists and cold-chain where the Distribution pillar lags a strong overall score.

CLOSE THE LAST-MILE GAP

Sequenced down the MAI, a launch pays back in ~5 months at 2.3x scripts per rep versus a blanket rollout.

THE FOUR-MOVE PLAY

- 01

Filter

Re-score with the Demand pillar (chronic prevalence) up-weighted for a diabetes lens.
- 02

Select

Take the top 40 districts on the diabetes-adjusted MAI as the launch wave.
- 03

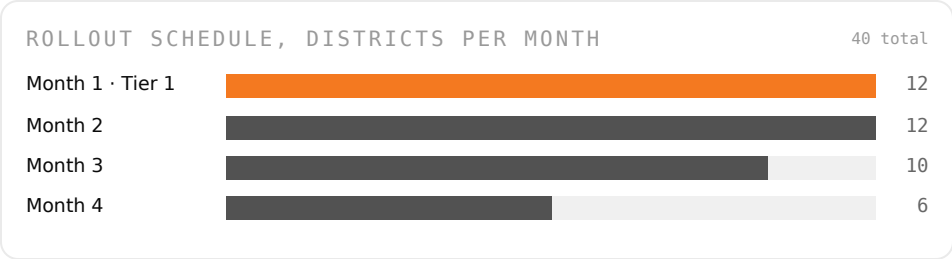
Sequence

12 Tier-1 districts in month 1, next 28 across months 2-4 as reps ramp.
- 04

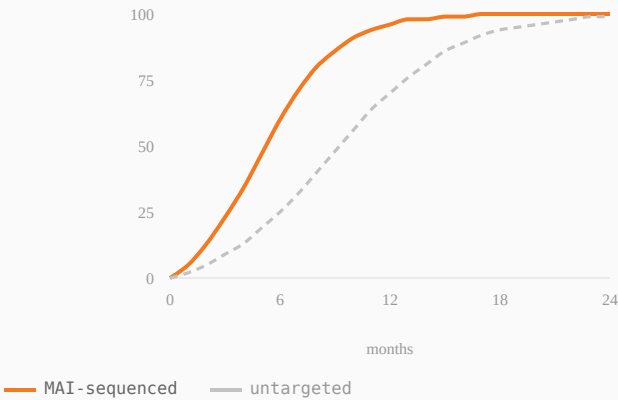
Resource

Match rep count and stockists to each district's Access and Distribution pillars.

Same budget, concentrated on measured readiness. The uplift comes from not spending the first rupees in districts that were never going to convert.



ADOPTION, BASS DIFFUSION MODEL



Targeting lifts the innovation (p) and imitation (q) coefficients, steepening the curve to faster peak adoption.



PROJECTED VS UNTARGETED

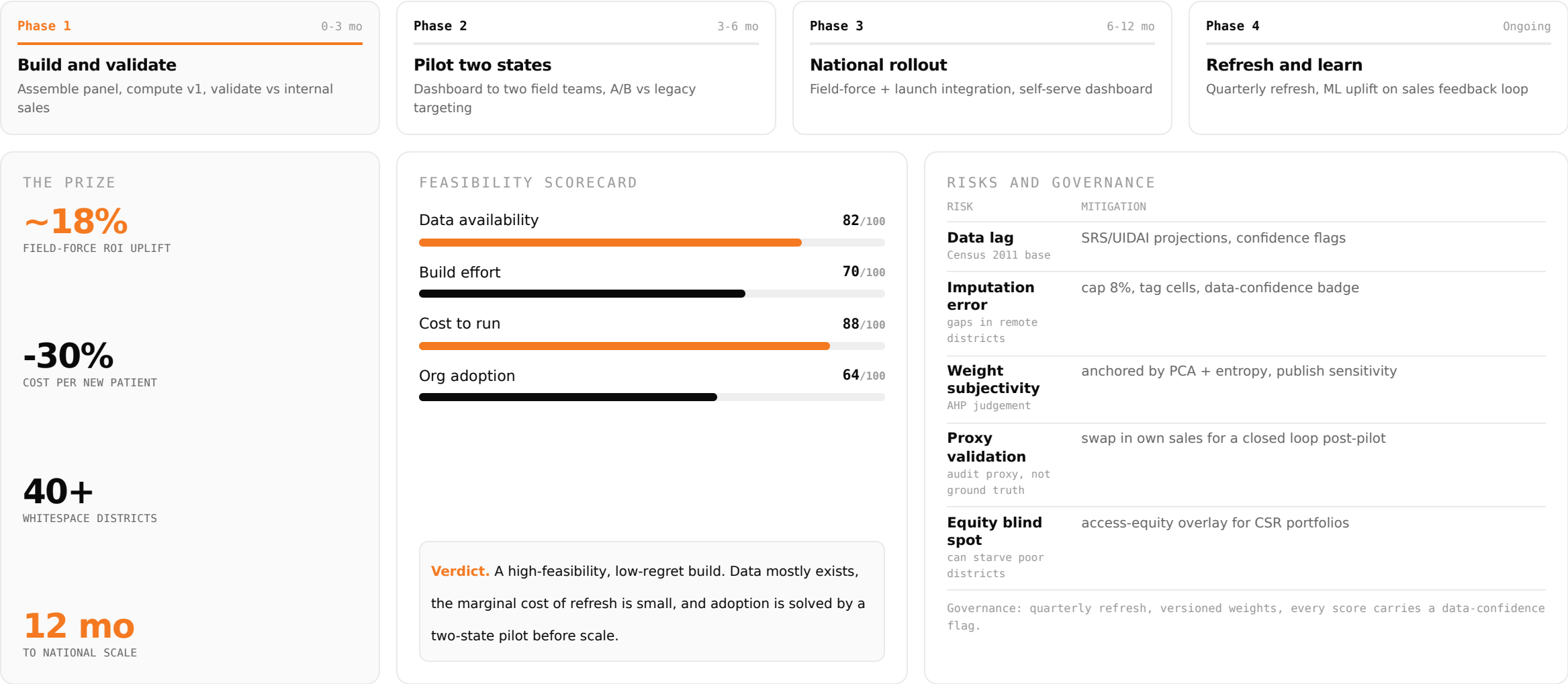
40
LAUNCH-WAVE DISTRICTS

2.3x
SCRIPTS PER REP

-31%
COST PER NEW PATIENT

~5 mo
TO PAYBACK

From a validated model to a national, self-refreshing commercial system within a year.



DISTRICT MARKET ATTRACTIVENESS INDEX

One transparent score, 730+ districts, every pharma decision pointed the same way.

Model

PCA + ENTROPY + AHP

Framework

GE-MCKINSEY, PORTER, BCG

Activation

4 COMMERCIAL USE CASES

Thank you · Annexure follows: frameworks, indicator dictionary, sources, formulae, worked example, references

Every framework used, and exactly where it enters the model.

FRAMEWORK	ROLE	HOW IT IS USED
GE-McKinsey nine-box	Strategy	The MAI is the market-attractiveness axis, computed objectively so districts can be placed in invest / build / harvest / exit boxes.
Porter's Five Forces	Strategy	Rivalry (sales saturation), buyer and supplier power and substitutes shape the Whitespace pillar.
PESTEL	Macro scan	Political, economic, social, technological, environmental and legal drivers define which indicators enter the Demand and Growth pillars.
MCDA / composite indicators	Method	OECD/JRC ten-step method governs selection, normalization, weighting and aggregation into one index.
AHP (Saaty)	Weighting	Expert pairwise judgement sets part of the cross-pillar weights, with a consistency ratio below 0.1.
PCA + Shannon entropy	Weighting	Objective, data-driven weights from variance loadings and information dispersion, reconciled with AHP.
BCG growth-share	Portfolio	Tiers map to Stars, Cash Cows, Question Marks and Dogs to set the play per district group.
Ansoff matrix	Growth	Penetration, market development, product development and diversification guide portfolio moves by tier.
STP	Targeting	Segment districts by MAI and pillar profile, target Tier 1-2 and whitespace, position portfolio to the dominant need.
Bass diffusion	Forecasting	Models adoption of a new launch; MAI targeting raises the innovation (p) and imitation (q) coefficients.
Pareto principle	Prioritisation	Confirms opportunity concentration, the top districts hold a disproportionate share of the prize.

All indicators, direction and within-pillar weight.

CODE	INDICATOR	DIR	WT
D1	Chronic disease prevalence (diabetes, CVD, respiratory)	+	0.30
D2	Population aged 45+ (share)	+	0.22
D3	Inpatient + outpatient load per 1k	+	0.20
D4	Communicable disease incidence	+	0.15
D5	Total population (addressable base)	+	0.13
A1	District domestic product per capita	+	0.32
A2	Health insurance / PMJAY coverage	+	0.24
A3	Monthly per-capita consumption exp.	+	0.22
A4	Formal-sector employment share	+	0.12
A5	Out-of-pocket health spend capacity	+	0.10
H1	Registered doctors per 10k population	+	0.28
H2	Hospital beds per 10k population	+	0.24
H3	PHC / CHC / sub-centre density	+	0.20

CODE	INDICATOR	DIR	WT
H4	Specialist & super-specialist footprint	+	0.16
H5	Diagnostic-lab density	+	0.12
R1	Retail pharmacy density per 10k	+	0.34
R2	Stockist & C&F presence	+	0.26
R3	e-Pharmacy serviceability	+	0.22
R4	Cold-chain / logistics reach	+	0.18
W1	Existing pharma sales saturation (inverse)	inverse	0.46
W2	Med-rep coverage intensity (inverse)	inverse	0.30
W3	Per-capita script gap vs peers	+	0.24
G1	Urbanization rate (delta)	+	0.34
G2	District income CAGR	+	0.34
G3	Population growth rate	+	0.18
G4	New-facility & construction permits	+	0.14

Real public datasets behind each pillar, with vintage and granularity.

P1 Demand

NFHS-5	2019-21 · District
Census / SRS projections	2011 / 2024 · District
HMIS (MoHFW)	2023-24 · District
IDSP / NCDC	2023 · District
Census / UIDAI proj.	2024 · District

P2 Affordability

State DES / District GDP	2022-23 · District
NHA (PMJAY), NFHS-5	2023 · District
HCES (MoSPI)	2022-23 · District/NSS region
EPFO / PLFS	2023-24 · District/State
NSSO Health Round	2017-18 · NSS region

P3 Access

NMC / State registries	2023 · District
RHS, HMIS, NIN	2023 · District
Rural Health Statistics	2022-23 · District
Practo/IMA panels (proxy)	2024 · District
NABL directory	2024 · District

P4 Distribution

State Drug Control / AIOCD	2024 · District
AIOCD-AWACS (proxy)	2024 · District
Platform pincode coverage	2024 · Pincode -> District
CDSCO, logistics indices	2023 · District

P5 Whitespace

AIOCD-AWACS retail audit (proxy)	2024 · District
Field-force benchmarks	2024 · District
Derived (demand vs sales)	2024 · District

P6 Growth

Census, town projections	2011-24 · District
State DES time series	2018-23 · District
SRS / projections	2011-24 · District
ULB / state records (proxy)	2023 · District

The formulae behind normalization, weighting and aggregation.

1 · NORMALIZATION (MIN-MAX)

$$x'_i = 100 \cdot (x_i - \min_i) / (\max_i - \min_i)$$

Negative indicators use (max - x) so higher is always better. Bounds set after winsorizing at [p5, p95].

2 · ENTROPY WEIGHT

$$e_i = -k \sum p_{ij} \ln p_{ij}$$
$$w^{ent}_i = (1 - e_i) / \sum (1 - e)$$

Indicators that discriminate more between districts earn more weight.

3 · RECONCILED WEIGHT

$$w_i = 0.4 \hat{w}^{pca} + 0.3 \hat{w}^{entropy} + 0.3 \hat{w}^{ahp}$$

PCA 40%

Entropy 30%

AHP 30%

4 · AGGREGATION

$$Pillar_p = \sum_{i \in p} w_i x'_i$$

MAI = $\prod_p Pillar_p^{w_p}$

Geometric across pillars penalizes imbalance; arithmetic within a pillar keeps it interpretable.

How Mumbai Suburban reaches its composite score.

PILLAR	SCORE	W	CONTRIBUTION
P1 Demand	81.2	0.24	2.87
P2 Affordability	78.7	0.21	2.50
P3 Access	79.3	0.20	2.40
P4 Distribution	80.2	0.15	1.93
P5 Whitespace	80.9	0.12	1.69
P6 Growth	79.9	0.08	1.42
Composite MAI (geometric)			80.0

READING IT

Each pillar score is raised to its cross-pillar weight and the results are multiplied. Because the aggregation is geometric, a weak pillar drags the composite down more than a simple average would, so only genuinely well-rounded districts reach the top tier.

$MAI = \prod \text{Pillar}_p^{W_p} = 80.0$

Methodology and data sources.

01

OECD / JRC. Handbook on Constructing Composite Indicators: Methodology and User Guide. 2008.

02

Census of India, Office of the Registrar General. District-level demographic tables, 2011 + SRS projections.

03

IIPS. National Family Health Survey (NFHS-5), 2019-21. District fact sheets.

04

MoHFW. Health Management Information System (HMIS) and Rural Health Statistics, 2022-24.

05

National Health Authority. PM-JAY beneficiary and coverage data, 2023.

06

MoSPI. Household Consumption Expenditure Survey (HCES), 2022-23; PLFS, 2023-24.

07

AIOCD-AWACS. Retail pharma audit (used as an illustrative validation proxy).

08

IDSP / NCDC, NMC registries, CDSCO, and State Directorates of Economics and Statistics.

09

Diakoulaki et al. Determining objective weights: the CRITIC / entropy method. 1995.

10

Saaty, T. The Analytic Hierarchy Process. 1980.

NOTE ON DATA

District scores in this deck are illustrative, generated to demonstrate the model's outputs and interfaces. The listed sources are real and district-resolvable; production scoring ingests them directly.